

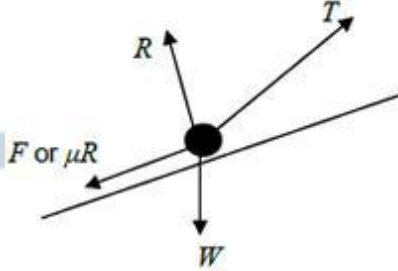
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Sums the forces given correctly	AO1.1b	B1	$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$ $= 33\mathbf{i} - 11\mathbf{j}$
	Uses Pythagoras to find the magnitude of the vector and obtains correct magnitude (given to 3 sig figs)	AO1.1b	B1	$ \mathbf{F} = \sqrt{33^2 + (-11)^2}$ $= 34.8 \text{ N (3 sf)}$
(ii)	Uses trig expression with appropriate values	AO1.1a	M1	$\tan \theta = \frac{11}{33}$
	Obtains correct angle (given to nearest 0.1°)	AO1.1b	A1	$\theta = \tan^{-1}\left(\frac{1}{3}\right)$ $= 18.4^\circ \text{ (3 sf)}$ OR $\sin \theta = \frac{11}{\sqrt{1210}}$ $\theta = \sin^{-1}\left(\frac{11}{\sqrt{1210}}\right)$ $= 18.4^\circ \text{ (3 sf)}$ OR $\cos \theta = \frac{33}{\sqrt{1210}}$ $\theta = \cos^{-1}\left(\frac{33}{\sqrt{1210}}\right)$ $= 18.4^\circ \text{ (3 sf)}$
(b)	States negative of 'their' part (a)(i)	AO2.2a	B1F	$\mathbf{F}_4 = -33\mathbf{i} + 11\mathbf{j}$
				Total 5 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
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(a)	Resolves horizontally and vertically to obtain expressions for F and R (allow consistent mixing of sin and cos)	AO3.4	M1	Resolving vertically $R = 8 \times 9.8 - 50\sin 40^\circ$ Resolving horizontally $F = 50\cos 40^\circ$
	Obtains correct expressions for R and F	AO1.1b	A1	
	States friction model $F = \mu R$ with 'their' values for F and R	AO1.2	B1	$F = \mu R$ $50\cos 40^\circ = \mu(8 \times 9.8 - 50\sin 40^\circ)$ $\mu = \frac{50\cos 40^\circ}{8 \times 9.8 - 50\sin 40^\circ}$
	Completes a rigorous argument that results with correct μ AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$\mu = 0.8279660445 = 0.83$ (2sf) AG
(b)	Draws correct diagram with exactly four forces showing arrow heads and labels Can use Mg or $8g$ or 78.4 for W	AO3.3	B1	
(ii)	Resolves perpendicular to the plane resulting in a three term equation containing; R , $8g\cos 5^\circ$ and $T\sin 40^\circ$ (or $T\cos 50^\circ$) OR Resolves horizontally and vertically to obtain equations of motion in horizontal and vertical directions	AO3.1b	M1	$R = 8 \times 9.8\cos 5^\circ - T\sin 40^\circ$ $= 78.1 - T\sin 40^\circ$
	Obtains correct expression for R OR Obtains correct horizontal and vertical equations	AO1.1b	A1	$T\cos 40^\circ - 8 \times 9.8\sin 5^\circ - F = 8 \times 3$ $T\cos 40^\circ - 8 \times 9.8\sin 5^\circ - 0.827... \times (8 \times 9.8\cos 5^\circ - T\sin 40^\circ) = 24$

Forms a four term equation of motion parallel to the plane with correct terms (allow sign errors) OR Eliminates R to solve for T	AO3.1b	M1	$T = \frac{24 - 8 \times 9.8 \sin 5^\circ + 0.827 \dots \times 8 \times 9.8 \cos 5^\circ}{\cos 40^\circ + 0.827 \dots \times \sin 40^\circ}$ $T = 73.55939193 = 74 \text{ (2 sf)}$
Obtains correct equation of motion OR Obtains correct expression(s) without R	AO1.1b	A1	
Uses the friction model in the four term equation of motion where R is in the form $a - bT$ and a and b are positive constants OR Uses the friction model in the horizontal and vertical equations Dependent on previous M1	AO3.1b	dM1	
Solves for T	AO1.1b	A1	
Obtains correct value of T with 2 sf accuracy. FT incorrect value found for T provided both M1 marks and dM1 mark have been awarded	AO3.2a	A1F	
			Total 12 marks

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Q3.

(a)

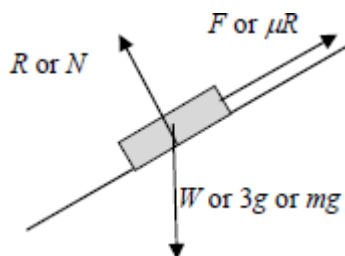


Diagram with exactly three forces showing arrow heads and labelled. If components are also shown they must use a different style e.g. dashed lines then they can be ignored. Friction must be up the slope.

B1

1

(b) $(R =) 3 \times 9.8 \cos 40^\circ = 22.5 \text{ N}$

M1: Resolving perpendicular to the slope. Can use $\sin 40^\circ$ or $\cos 50^\circ$ for method mark, with g or 9.8 .

*A1: Correct normal reaction. Accept
AWRT 22.5 (Note use of 9.81 still gives 22.5 N.)*

M1A1

2

(c) $(F =) 0.2R = 4.50 \text{ N}$

M1: Use of $F = \mu R$.

A1F: Correct friction. Accept 4.5 N or AWRT 4.50.

(Accept 4.51 N from the use of 9.81.)

Follow through incorrect normal reaction from part (b).

M1A1F

2

(d) $3a = 3 \times 9.8 \sin 40^\circ - 4.504$

M1: Three term equation of motion with correct terms, with $3a$, either component of weight and their answer to part (c) for F .

A1F: Equation of motion with correct terms and signs.

M1A1F

$a = 4.80 \text{ m s}^{-2}$

A1F: Correct acceleration. Accept 4.8 or AWRT 4.80.

(Note that using 9.81 still gives 4.80 m s^{-2}).

Follow through friction from part (c).

A1F

3

(e) No air resistance force acting

or

No other forces acting on the box.

or

They (forces in the diagram) are the only forces that act.

OR

No turning effect (due to forces).

or

Forces are concurrent. OE

B1: Correct assumption.

Ignore irrelevant comments

B1

1

[9]

Q4.

(a) $F^2 = 70^2 + 40^2 - 2 \times 40 \times 70 \cos 150^\circ$

M1: Use of cosine rule with an obtuse angle and a – sign.

M1

A1: Correct expression.

A1

$$F = \sqrt{11350}$$

dm1: Taking square root of a value > 6500. May be implied by final answer.

dm1

$$F = 107$$

A1: Correct resultant to 3sf or more.

Accept AWRT 106 or 107.

A1

4

OR by components,

$$70 + 40 \cos 30^\circ (= 104.64)$$

M1: Finding two perpendicular components of the resultant, with same force (usually the 40 N force) resolved in both expressions.

Allow consistent sin / cos confusion.

A1: Both components correct.

(Note that resolving parallel and perpendicular to the 40 N force gives components of 100.6 and 35)

(M1A1)

$$40 \sin 30^\circ (= 20)$$

dm1: Finding the magnitude of the resultant.

(dm1)

$$F = \sqrt{104.64^2 + 20^2} = 107 \text{ N}$$

A1: Correct resultant to 3sf or more.

Accept AWRT 106 or 107.

(A1)

(4)

(b) $\frac{\sin \alpha}{40} = \frac{\sin 150^\circ}{106.54}$

M1: Use of sine rule with 150°, their answer to part (a) and 40 or 70.

A1: Correct equation with AWRT 106 or 107.

M1A1

$$\alpha = 10.8^\circ$$

A1: Correct angle. Accept 10.9°.

A1

3

OR

$$\tan \alpha = \frac{20}{104.64}$$

M1: Use of tan with 20 and AWRT 104 or 105.

A1: Expression for tan α in the form

$$\tan \alpha = \frac{20}{\text{AWRT } 104 \text{ or } 105}.$$

Could be implied by their final answer.

(M1A1)

$$\alpha = 10.8^\circ$$

A1: Correct angle. Accept 10.9°.

(A1)

(3)

OR

$$\sin \alpha = \frac{20}{106.53}$$

M1: Use of sin with 20 or AWRT 104 or 105 in the numerator and their answer to (a) as the denominator.

A1: Expression for sin α in the form

$$\sin \alpha = \frac{20}{\text{AWRT } 106 \text{ or } 107}.$$

(M1A1)

$$\alpha = 10.8^\circ$$

A1: Correct angle. Accept 10.9°.

(A1)

(3)

OR

$$\cos \alpha = \frac{104.64}{106.53}$$

M1: Use of cos with 20 or AWRT 104 or 105 in the

numerator and their answer to (a) as the denominator.

A1: Expression for $\cos \alpha$ in the form

$$\cos \alpha = \frac{\text{AWRT } 104 \text{ or } 105}{\text{AWRT } 106 \text{ or } 107}.$$

(M1A1)

$$\alpha = 10.8^\circ$$

A1: Correct angle. Accept 10.9° or 10.7° .

Apply ISW if $180^\circ - \alpha$ is seen after finding α .

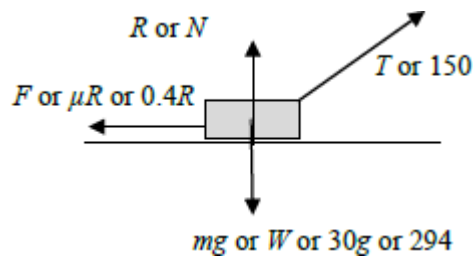
(A1)

(3)

[7]

Q5.

(a) (i)



B2: Correct diagram with exactly four forces showing arrow heads and labelled.

B1: Diagram with one error or omission.

B0: Diagram with 2 or more errors or omissions.

If components are also shown and they use a different style, eg dashed lines, they can be ignored.

If both components are shown in the same style as other forces, this counts as two errors.

Note; Do not accept 30kg for the weight.

B1

B1

2

(ii) $R + 150 \sin 20^\circ = 30 \times 9.8$

$$(R =) 30 \times 9.8 - 150 \sin 20^\circ$$

$$= 242.69 \dots$$

M1: Resolving vertically to obtain a three term equation, with R , $150 \sin$ or $\cos(20^\circ \text{ or } 70^\circ)$ and $30g$ oe.

A1: Correct equation. Allow g instead of 9.8 .

M1A1

= 243 N (to 3sf)

A1: AG Correct final answer having seen either 2nd or 3rd or both line of solution.

A1

3

(iii) ($F =$) $0.4 \times 242.7 = 97.1 \text{ N}$

M1: Use of $F = \mu R$ or $F \leq \mu R$.

A1: Correct final answer without an inequality.

Accept 97.2.

M1A1

2

(iv) $30a = 150 \cos 20^\circ - 97.08$

M1: Three term equation of motion with $30a$, $150 \sin$ or $\cos(20 \text{ or } 70^\circ)$ and their friction from (a)(iii). Condone incorrect signs.

A1: Correct equation.

M1A1

$$a = \frac{150 \cos 20^\circ - 97.08}{30} = 1.46 \text{ m s}^{-2}$$

dM1: Solving for a .

dM1

A1: Correct acceleration. Accept 1.45 or 1.47 or AWRT 1.46.

A1

4

(b) $R = 30 \times 9.8 - T \sin 20^\circ$

B1: Correct normal reaction in terms of T .

B1

$$F = 0.4 (30 \times 9.8 - T \sin 20^\circ)$$

B1: Correct friction in terms of T

B1

$$T \cos 20^\circ = 0.4 (30 \times 9.8 - T \sin 20^\circ)$$

M1: Resolving tension horizontally and equating to F , provided that F is in terms of T .

A1: Correct equation.

M1A1

$$T = \frac{0.4 \times 30 \times 9.8}{\cos 20^\circ + 0.4 \sin 20^\circ} = 109 \text{ N}$$

A1: Correct tension. AWR T 109.

A1

5

(c) The same

B1: The same.

Use of $g = 9.81$ gives acceptable final answers.

B1

1

[17]

Q6.

(a) (i) $10^2 = 4^2 + 2 \times a \times 50$

M1: Use of a constant acceleration equation to find a , with v and u substituted correctly.

For example $4^2 = 10^2 + 100a$ scores M0A0A0.

A1: Correct constant acceleration equation.

M1A1

$$a = \frac{100 - 16}{100} = 0.84 \text{ ms}^{-2}$$

A1: Correct a .

A1

Note if t found first award M1 for use of

$$v = u + at \text{ or } s = ut + \frac{1}{2} at^2.$$

3

(ii) $50 = \frac{1}{2} (4 + 10)t$

M1: Use of a constant acceleration equation to find t .

A1F: Correct constant acceleration equation with their acceleration from (a)(i) seen.

M1A1

$$t = \frac{50}{7} = 7.14 \text{ s}$$

A1: Correct t . Accept $\frac{50}{7}$ or $7\frac{1}{7}$ or

AWRT 7.14.

A1

OR

$$10 = 4 + 0.84t$$

If t has been found in part (a) the working does not have to be repeated, but value of t must be stated.

(M1A1F)

$$t = \frac{6}{0.84} = 7.14 \text{ s}$$

Do not follow through incorrect values of a .

(A1)

OR

$$50 = 4t + \frac{1}{2} \times 0.84t^2$$

(M1A1F)

$$0.42t^2 + 4t - 50 = 0$$

$$t = 7.14 \text{ (or } t = -16.6)$$

(A1)

3

(b) $70 \times 0.84 = 58.8 \text{ N}$

M1: Use of $F = ma$ with $m = 70$ and their a from (a)(i).

A1F: Correct F . Follow through their value of a from part (a)(i).

M1A1F

2

(c) (i) $58.8 = 70 \times 9.8 \sin \alpha$

$$\sin \alpha = \frac{58.8}{70 \times 9.8} = 0.08571$$

M1: Resolving parallel to the slope must see $70g$ or mg OE with $\sin \alpha$ or $\cos \alpha$ and their answer to part (b).

A1F: Correct equation. Follow through their answer to part (b) provided $\sin \alpha < 1$

M1A1F

$$\alpha = 4.92^\circ$$

A1F: Correct angle. Follow through their answer to part (b). Accept 4.91° provided $\sin \alpha < 1$.

A1F

3

(ii) $70 \times 9.8 \sin \alpha - 30 = 58.8$

$$\sin \alpha = 0.12945$$

M1: Three term equation of motion. Must see $70g$ or mg OE with $\sin \alpha$ or $\cos \alpha$.

A1F: Correct equation. Follow through their answer to part (b) provided $\sin \alpha < 1$

M1A1F

$$\alpha = 7.44^\circ$$

A1F: Correct angle. Follow through their answer to part (b) provided $\sin \alpha < 1$.

Accept 7.43° .

Accept 7.41° from 0.129.

A1F

3

- (d) The air resistance force will increase (vary or change) with speed.

B1: Correct statement.

B1

1

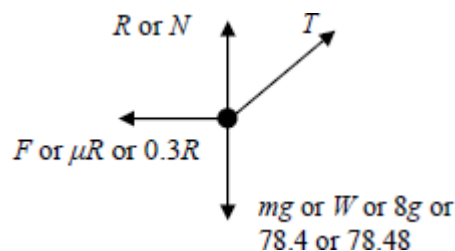
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[15]

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Q7.

(a)



*B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.*

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or $0.3R$ for F .

B1

1

(b) $R + 40 \sin \theta = 8 \times 9.8$

M1: Resolving vertically to obtain a three term equation, with R , $T \sin$ or $\cos(30^\circ$ or $60^\circ)$ and $8g$ oe.

A1: Correct equation

M1A1

$$R = 78.4 - 40 \sin \theta$$

A1: Correct expression for R .

Accept $(R=)8g - T \sin 30^\circ$

Note if using $g = 9.81$ accept

$R = 78.48 - 0.5T$ or $R = 78.5 - 0.5T$

A1

3

(c) $8a = 40 \cos \theta - 0.3(78.4 - 40 \sin \theta)$

M1: Use of the friction inequality with R from part (b)

A1: Correct equation.

M1A1

$$8a = -23.52 + 40 \cos \theta + 12 \sin \theta$$

M1: Use of equation of motion with $40 \sin \theta$ or $40 \cos \theta$ and $8a$ and a friction term.

M1

$p = -2.94$

$q = 8$

$r = 1.5$

A1: One correct value.

A1

A1: All three values correct.

A1

5

[9]

Q8.

(a) Using $F = ma$:

$$m \frac{dv}{dt} = 49 - 9.8v \quad \text{or} \quad 5g - 9.8v$$

Need to see $m \frac{dv}{dt}$ or $5 \frac{dv}{dt}$ or $a = \frac{49-9.81}{5}$

M1

$$\therefore \frac{dv}{dt} = -1.96(v - 5)$$

Must see m terms (not $a = \dots$)

A1

2

(b) $\int \frac{dv}{v-5} = -1.96 \int dt$

And one side integrated

M1

$$\ln(v - 5) = -1.96t + c$$

Need $+c$, A1 each side

A1A1

When $t = 0$, $v = 7 \Rightarrow c = \ln 2$

OE

A1

$$\ln \frac{v-5}{2} = -1.96t$$

$$\frac{v-5}{2} = e^{-1.96t}$$

$$v = 5 + 2e^{-1.96t}$$

CAO

A1

5

[7]

Q9.

(a) $P \cos 80^\circ - Q \cos 80^\circ = 250a$

M1: Horizontal equation of motion in the form

$$P \cos 80^\circ \pm Q \cos 80^\circ = 250a \quad \text{or}$$

$$P \sin 80^\circ \pm Q \sin 80^\circ = 250a$$

A1: Correct horizontal equation.

M1A1

$$P \sin 80^\circ + Q \sin 80^\circ = 250g$$

B1: Correct vertical equation.

Note: the above marks could be awarded for a correct vector equation.

B1

$$P - Q = \frac{250a}{\cos 80^\circ}$$

$$P + Q = \frac{250g}{\sin 80^\circ} \quad \text{AG}$$

$$2P = \frac{250a}{\cos 80^\circ} + \frac{250g}{\sin 80^\circ}$$

dM1: Solving for P with an attempt to eliminate Q.

dM1

$$P = 125 \left(\frac{a}{\cos 80^\circ} + \frac{g}{\sin 80^\circ} \right)$$

A1: Correct result from correct working.

Must see an expression for 2P or 2P sin 80° cos 80°

A1

5

(b) $P \cos 80^\circ = 250a$

M1: Using Q = 0 into correct original equation(s) or resolving without Q.

M1

$$P \sin 80^\circ = 250g$$

$$\frac{1}{\tan 80^\circ} = \frac{a}{g}$$

dM1: Eliminating P

dM1

$$a = \frac{g}{\tan 80^\circ} = 1.73$$

A1: Correct a.

Note: use of P = ± Q scores M0dM0A0

Note: use of P = 0 can lead to ± 1.73 but scores M0dM0A0 unless fully justified by a symmetry argument.

A1

3

[8]

Q10.

(a) $\mathbf{v} = 4.5\mathbf{i} + 2\mathbf{j}$

B1: Correct velocity

B1

1

(b) $\mathbf{v} = 4.5\mathbf{j}$

B1: Correct velocity

B1

1

(c) Due North

B1: Correct direction

B1

1

(d) $4.5\mathbf{j} = 4.5\mathbf{i} + 2\mathbf{j} + 5\mathbf{a}$

M1: Use of a constant acceleration equation to find a

A1: Correct equation

M1A1

$$\mathbf{a} = \frac{-4.5\mathbf{i} + 2.5\mathbf{j}}{5} = -0.9\mathbf{i} + 0.5\mathbf{j}$$

A1: Correct acceleration

A1

3

(e) $\mathbf{F} = 250(-0.9\mathbf{i} + 0.5\mathbf{j}) = -225\mathbf{i} + 125\mathbf{j}$

M1: Using Newton's second law.

M1

$$F = \sqrt{225^2 + 125^2} = 257 \text{ N}$$

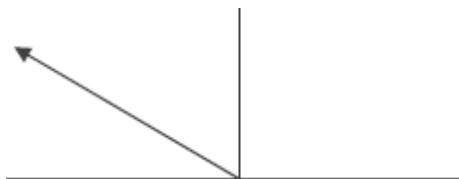
M1: Finding magnitude.

A1: Correct magnitude of the force

M1A1

3

(f)



B1: Diagram with arrow in the correct quadrant.

B1

$$\tan^{-1}\left(\frac{125}{225}\right) = 29.1^\circ$$

M1: Use of trig to find an angle.

M1

A1: Correct angle from working.

A1

OR

$$\tan^{-1}\left(\frac{0.5}{0.9}\right) = 29.1^\circ$$

$$180 - 29.1 = 151^\circ$$

A1: Finding required angle correctly.

A1

4

[13]

Q11.

(a) Resultant = $(6\mathbf{i} - 3\mathbf{j}) + (3\mathbf{i} + 15\mathbf{j})$
 $= 9\mathbf{i} + 12\mathbf{j}$

M1: Summing the two vectors

M1

A1: Correct resultant

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A1

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2

(b) Magnitude = $\sqrt{9^2 + 12^2}$
 $= 15 \text{ N}$

M1: Finding magnitude with an addition sign.

M1

A1F: Correct magnitude based on their answer to part (a).

A1F

2

(c) $1.5m = 9$ $2m = 12$
 or
 $m = 6 \text{ kg}$ $m = 6 \text{ kg}$

M1: Applying Newton's second law to one or both of the components.

M1

A1F: Correct mass, follow through their answer to part (a). Do not award this mark if vector division with 2 components has been used, eg

$$\frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6 \quad \text{or } 6\mathbf{i} + 6\mathbf{j} \text{ etc without a correct previous statement gives M0A0}$$

A1F

2

(d) (i) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j})t^2$

M1: Using a constant acceleration equation to find the position vector with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

A1: Correct position vector.

A1

2

(ii) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j}) \times 2^2 = 3\mathbf{i} + 4\mathbf{j}$
 $d = \sqrt{(3)^2 + (4)^2}$

M1: Finding the position vector when $t = 2$

($\mathbf{r} = (1.5\mathbf{i} + 2\mathbf{j}) \times 2 = 3\mathbf{i} + 4\mathbf{j}$ scores M0 unless it is clear how the 2 was obtained, possibly by a correct formula in (d) (i))

M1

$$= \sqrt{25} = 5$$

A1: Correct distance

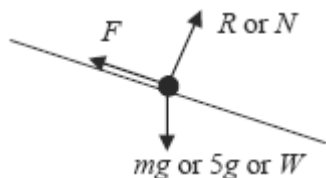
A1

2

[10]

Q12.

(a)



Correct force diagram with labels and arrows
Accept components of the weight if shown in a different notation with the weight also shown.
B0 if components are shown instead of the weight.

B1

1

(b) $(R =) 5 \times 9.8 \cos 40^\circ = 37.5 \text{ N}$ **AG**

Attempt at resolving perpendicular to the slope (eg $49 \sin 40^\circ$)

M1

Correct value from correct working

A1

2

(c) $5 \times 0.8 = 5 \times 9.8 \sin 40^\circ - \mu \times 5 \times 9.8 \cos 40^\circ$

Use of $F = \mu R$ at any stage and with any F but with $R = 37.5$ OE

B1

Three term equation of motion from resolving parallel to the slope with weight component, friction and ma term.

M1

Correct terms seen (may be as 31.5, 37.5μ (or F) and 4)

A1

Correct signs

A1

$$\mu = \frac{5 \times 9.8 \sin 40^\circ - 5 \times 0.8}{5 \times 9.8 \cos 40^\circ} = 0.733$$

Solving for μ

m1

A1: Correct value for μ

Allow 0.732 but not $\frac{11}{15}$ unless converted to a decimal

A1

6

- (d) There is less friction so the coefficient of friction must be less.

Less friction

B1

*Smaller coefficient of friction
If the answer and explanation contradict each other, award no marks*

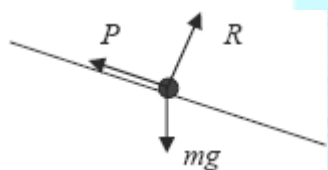
B1

2

[11]

Q13.

(a)



*Correct diagram with arrows and labels
Must not use F instead of P
Condone resistance instead of P*

B1

1

- (b) $P = 100 \times 9.8 \sin 4^\circ$

Resolving weight (must see 100)

M1

Using $\sin 4^\circ$ or $\cos 86^\circ$

M1

$= 68.4$

AG *Correct P from correct working*

A1

3

- (c) $100a = 100 \times 9.8 \sin 5^\circ - 100 \times 9.8 \sin 4^\circ$

Three term equation of motion

M1

Weight resolved correctly

A1

Correct equation

A1

$$a = \frac{100 \times 9.8 \sin 5^\circ - 100 \times 9.8 \sin 4^\circ}{100}$$

$$= 0.171$$

Correct a. (Accept 0.170 or 0.17)

A1

4

- (d) You would expect P to vary with the speed of the car.

Correct explanation

B1

1

[9]

Q14.

- (a) (i) $R = 80 \cos 25^\circ$
component attempted

M1

correct component

A1

$$R = 72.5\text{N}$$

CAO

A1

3

- (ii) $F = 0.32 \times 72.5$
condone inequality

M1

$$F = 23.2\text{N}$$

CAO

A1

2

- (iii) $T + F = 80 \cos 65^\circ$

3 forces direction correct, component

M2

attempted

A1

$$T = 10.6\text{N}$$

ft friction

A1ft

4

(iv) $T = F + 80\cos 65^\circ$

3 forces, direction correct, component

M1

attempted

A1

$$T = 57.0\text{N} (57\text{N})$$

ft friction

A1ft

3

(iv) $\text{Mass} = \frac{80}{g} = (8.16\text{kg})$

B1

1

(b) $80 \cos 65^\circ - F = \text{mass} \times \text{acceleration}$

3 terms, component attempted

M1

$$10.6 = \frac{80}{g} \times \text{acc}$$

all correct

A1

$$\text{acc} = 1.30 \text{ ms}^{-2} (1.3 \text{ ms}^{-2})$$

CAO

A1

3

[16]

Q15.

(a) $P = 5 + 8\cos 60^\circ$

Both relevant forces, component of 8N attempted

M1

All correct

A1

$P = 9$

CAO

A1

3

(b) $Q = 8\cos 30^\circ$

Component of 8N attempted

M1

$Q = 6.93$ or $4\sqrt{3}$

AWRT 6.93

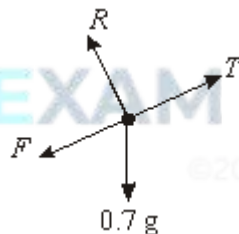
A1

2

[5]

Q16.

(a)



Accept W or mg (or 6.86) for weight

Arrows and labels needed

(can replace W with 2 correct components)

B1

1

(b) $R = 0.7g \cos 22^\circ$

component of weight attempted

M1

$R = 6.36 \text{ N}$

all correct, including signs

	CAO	A1	
		A1	3
(c)	$F = 0.25 \times 6.36$ $F = 1.59 \text{ N}$ CAO		

		M1 A1	2
(d)	$5.6 - 0.7g \sin 22^\circ$ $a = 2.06 \text{ ms}^{-2}$ <i>4 terms with weight component attempted</i>		

M1

A marks -1 each error, accept $\pm 0.7a$

A2

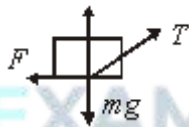
FT one error, accept $\pm$

A1F

[10]

Q17.

(a)



Correct diagram

B1

1

(b)	$40 \cos 30^\circ - F = 25 \times 0.1$ $F = 40 \cos 30^\circ - 2.5 = 32.1 \text{ N}$ <i>Three term equation of motion</i>		
-----	---	--	--

Correct equation

M1

A1

AG; correct force from correct working

A1

3

(c)	$R + 40 \sin 30^\circ = 25 \times 9.8$		
-----	--	--	--

$$R = 225 \text{ N}$$

Resolving vertically

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(d) $32.1 = 225\mu$

use of $F = \mu R$

M1

$$\mu = \frac{32.1}{225} = 0.143$$

Correct μ

A1

2

[9]

Q18.

(a)



Correct diagram

B1

1

(b) $40 \cos 30^\circ - F = 25 \times 0.1$

$$F = 40 \cos 30^\circ - 2.5 + 32.1 \text{ N}$$

Three term equation of motion

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(c) $R + 40 \sin 30^\circ = 25 \times 9.8$

$R = 225 \text{ N}$

Resolving vertically

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(d) $32.1 = 225\mu$

use of $F = \mu R$

M1

$\mu = \frac{32.1}{225} = 0.143$

Correct μ

A1

2

(e) Friction will decrease as normal reaction

Decrease in friction

B1

decreases

Normal reaction decreases

B1

2

[11]

Q19.

(a) $200 \sin 30^\circ = T \sin 45^\circ$

$T = \frac{200 \sin 30^\circ}{\sin 45^\circ} = 141 \text{ N}$

Resolving horizontally

M1

Correct equation

A1

AG; correct T from correct working

A1

3

(b) $200 \cos 30^\circ + 141 \cos 45^\circ + R = 500 \times 9.8$
Resolving vertically with four terms

M1

Correct values

A1

Correct signs

A1

$R = 4630 \text{ N}$

Correct R

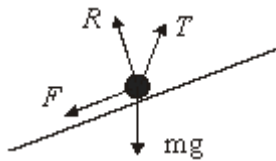
A1

4

[7]

Q20.

(a)



B1: Correct force diagram

B1

1

(b) $R + 20 \sin 30^\circ = 6g \cos 10^\circ$

M1: Resolving perpendicular to the slope with 3 terms

A1: Correct equation

M1A1

$R = 6g \cos 10^\circ - 20 \sin 30^\circ$

dM1: Solving for R

dM1

$R = 47.9 \text{ N (to 3 sf)}$ AG

A1: Correct R from correct working

A1

4

(c) $F = \mu R$

M1: Use of $F = \mu R$

M1

$$6 \times 0.4 = 20 \cos 30^\circ - 6g \sin 10^\circ - \mu R$$

M1: Resolving parallel to slope to get 4 term equation of motion

A1: Correct equation

M1A1

$$\mu R = 4.710$$

A1: Correct $F / \mu R$

A1

$$\mu = \frac{4.710}{47.91} = 0.0983$$

dM1: Solving for μ

dM1

A1: AWR 0.098

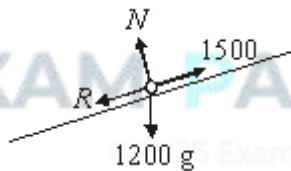
A1

6

[11]

Q21.

(a) (i)



3 correct & labelled

B1

All correct & labelled, no extras

B1

2

Ignore additional components of weight

(ii) $1500 = R + 1200 \times 9.8 \times \frac{1}{14}$

*M1: 3 terms with component attempted
($\alpha = 4.096$) – 1 each error*

M1A2

$$R = 660$$

AWRT

A1
4

(iii) 15 ms^{-1}

B1
1

(b) (i) $\frac{1500}{-1200}$

Accept $\pm$

M1

Mag of retardation = 1.25 ms^{-2}

A1
2

(ii) $0 = 15 + (-1.25) t$

velocities correct, but accept $\pm$ acceleration

M1

$t = 27.3 \text{ sec}$ (27.27)

ft one slip

A1F
2

[11]

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Q22.

(a) $1200 - R = 1000 \times 0.25$

M1: all relevant terms used

M1A1

$R = 950 \text{ N}$

f.t. one slip if $R > 0$

A1F
3

(b) $2100 - 1000g \times 0.1 - 950 = 1000 \times a$

M1: 3 or 4 terms considered

M1

-1 each term incorrect (any error),

or missing, $\alpha = 5.74^\circ$

A1A1

$$a = 0.17 \text{ ms}^{-2}$$

f.t. one error if 4 terms considered

A1F

4

[7]

Q23.

(a) (i) $T = 0.6a$

Either equation (M1 A0 for use of $0.1ga$)

M1

$$0.1g - T = 0.1a$$

SC who;e strng method:

A1

$$0.1g = 0.7a \text{ M1A1 (total mass used)}$$

A1

$$0.1g = 0.7a$$

$$a = 1.4 \text{ A1, max3/5}$$

m1

$$a = 1.4 \text{ ms}^{-2}$$

A1

5

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(ii) $T = 0.6 \times 1.4 = 0.84 \text{ N}$

Dependent on M1 gained in (a),

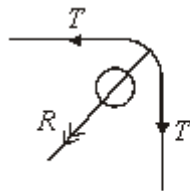
Or, s.c. can gain M1 (from (a)) A1 here if equations involving T not found in (a)

Max M1 A1

A1

1

(iii)



$$R = 2T \cos 45^\circ$$

M1

recognising 2 tensions involved

B1

M1: attempt at Pythagoras or at a component of T

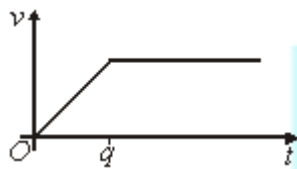
$$= 1.19 \text{ N}$$

A1: f.t. tension

A1F

3

(b)



1st line sloping and through O

B1

2nd line horizontal

B1

label at t q

B1

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3

[12]

Q24.

- (a) Sum of vertical forces = zero
used

M1

$$4\cos 60^\circ + 8\cos 60^\circ - q = 0$$

M1 for equation, 3 forces

M1 A1

$$q = 6$$

cao

A1

4

(b) horizontal:

$$R = 3 + 8 \cos 30^\circ - 4 \cos 30^\circ$$

M1A1

$$R = 6.46 \text{ or } 6.47$$

A1F

3

[7]

Q25.

(a) (i) $B: T = mg = 0.1 \times 9.8 = 0.98 \text{ N}$
cao

B1

1

(ii) $A: F = T (= 0.98) \quad R = 0.5 \times 9.8$

B1

$$\mu \times 0.5 \times 9.8 = 0.98$$

Condone inequality

M1

$$\mu = 0.2$$

cao

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A1

3

(iii)

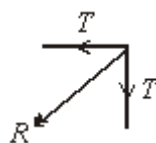


Must label T or 0.98 N

B1

1

(iv)



$$R = \sqrt{T^2 + T^2} \text{ or } 2T \cos 45^\circ$$

M1

$$R = 1.39\text{N}$$

A1

2

- (b) (i) $A: T - 0.98 = 0.5a$
Either equation for M1

M1A1

$$B: 2 \times 0.1 \times 9.8 - T = 0.2a$$

A1

$$0.98 = 0.7a$$

m1

$$a = 1.4$$

cao

A1

5

- (ii) $T = 0.98 + 0.5 \times 1.4, \quad T = 1.68$
ft accuracy error in equation

A1F

1

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(iii) $s = ut + \frac{1}{2}at^2$

$$0.7 = 0 + \frac{1}{2} \times 1.4t^2$$

M1

$$t = 1$$

A1

2

[15]

Q26.

- (a) $6 = a \cos 60^\circ + 8 \cos 60^\circ$
3 terms & attempt at components for M1 correct

components

M1

$$6 = \frac{a}{2} + 4$$

signs correct

A1A1

$$a = 4$$

A1

4

(b) $\pm(8\cos 30^\circ - a\cos 30^\circ)$

Components parallel to Oy only for M1
signs correct, 8 and a (or 4) used

M1A1

$$3.46 \text{ or } 3.47$$

ft one error e.g. sign or incorrect a

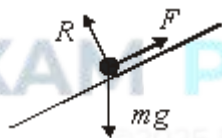
A1F

3

[7]

Q27.

(a) (i)



Correct force diagram

B1

1

(ii) $R = 20 \times 9.8 \cos 40^\circ = 150 \text{ N}$

Resolving perpendicular to the slope

M1

ag Correct answer from correct working

A1

2

(iii) $F = 20 \times 9.8 \sin 40^\circ = 126 \text{ N}$

Resolving parallel to the slope

M1

$$126 = \mu \times 150$$

Correct friction

A1

$$\mu = \frac{126}{150} = 0.84$$

Use of $F = \mu R$

M1

ag *Correct μ from correct working*

A1

4

- (b) It would be reduced
Correct explanation

B1

1

[8]

Q28.

- (a) $150g - T = 150 a$
Three term equation of motion for tank.

M1

$$T - 200g \sin 30^\circ = 200 a$$

Correct equation

A1

$$150g - 200a - 200g \sin 30^\circ = 150 a$$

Three term equation of motion for trolley

M1

Correct equation

A1

$$a = \frac{150g - 200g \sin 30^\circ}{350} = 1.4 \text{ ms}^{-2}$$

Solving for a

m1

Correct equation for a

A1

ag Correct a from correct working.

A1

7

(b) $T = mg$

Equations of motion for both bodies.

M1

$$T = 200 \text{ g} \sin 30^\circ$$

Correct equation for tank.

A1

Correct equation for trolley

A1

$$m = 200 \sin 30^\circ = 100 \text{ kg}$$

Correct mass

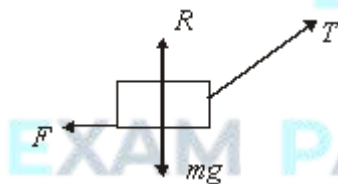
A1

4

[11]

Q29.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 30^\circ = 200 \times 9.8$

Three term equation from resolving vertically

M1

$$R + 0.5 T = 1960$$

$$R = 1960 - 0.5 T$$

Correct equation

A1

ag Correct result from correct working

A1

3

(c) $F = T \cos 30^\circ$

Resolving horizontally

M1

$$T \cos 30^\circ = 0.6(1960 - 0.5T)$$

Use of $F = \mu R$

M1

$$T (\cos 30^\circ + 0.3) = 1176$$

Correct equation

A1

Solving for T

m1

$$T = \frac{1176}{(\cos 30^\circ + 0.3)} = 1010 \text{ N}$$

Correct T

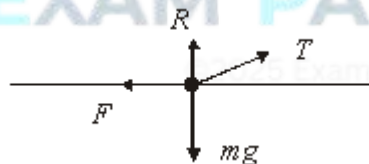
A1

5

[9]

Q30.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 40^\circ = 50 \times 9.8$

$$R = 490 - T \sin 40^\circ$$

Three term equation of motion

M1

Correct equation

A1

Correct expression for R
 Use of $F R = \mu$

A1

3

(c) $F = 0.6 (490 - T \sin 40^\circ)$

Use of $F = \mu R$

M1

$= 294 - 0.6 T \sin 40^\circ$

ag Correct result from correct working

A1

2

(d) $T \cos 40^\circ - (294 - 0.6 T \sin 40^\circ) = 50 \times 0.5$

$$T = \frac{319}{\cos 40^\circ + 0.6 \sin 40^\circ} = 277 \text{ N}$$

Four term equation of motion

M1

Correct equation

A1

Solving for T

M1

Correct T

A1

4

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[10]

Q31.

(a) $R + 100 \sin 20^\circ + 60 \sin 50^\circ = 490$

Resolving vertically

Correct equation

M1 A1

$R = 410$

A1

3

(b) (i) $100 \cos 20^\circ - 60 \cos 50^\circ - F = 25$

Four term equation of motion

Correct equation

M1 A1

$$F = 30.4$$

Solving for F

Correct F from from correct working

M1 A1

4

(ii) $30.40 = \mu \times 409.8$

Use of $F = \mu R$

M1

$$\mu = 0.0742$$

Correct μ

A1

2

[9]

Q32.

(a) $T_1 = 4 \times 9.8 = 39.2 \text{ N}$

Correct tension in left string

B1

$$T_2 = 3 \times 9.8 = 29.4 \text{ N}$$

Correct tension in right string

B1

2

(b) $39.2 \sin 43^\circ = 29.4 \sin \theta$

Resolving horizontally

M1

$$\sin \theta = \frac{39.2 \sin 43^\circ}{29.4} = 0.9093$$

Correct equation

A1

Correct expression for $\sin \theta$

A1

$$\theta = 65.4^\circ$$

Finding θ

M1

ag *Correct θ from correct working*

A1

5

(c) $9.8m = 39.2 \cos 43^\circ + 29.4 \cos \theta$

Resolving vertically

M1

Correct equation

A1

$$m = \frac{39.2 \cos 43^\circ + 29.4 \cos \theta}{9.8} = 4.17 \text{ kg}$$

Finding m

M1

Correct m

A1

4

[11]

Q33.

(a) $5^2 = 1^2 + 2 \times 4a$

Using constant acceleration equation to find a

M1 A1

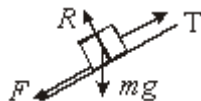
$a = 3 \text{ ms}^{-2}$

Correct equation

A1

3

(b) (i)



Correct force diagram

B1

1

(ii) $R = 20 \times 9.8 \cos 30^\circ = 170 \text{ N}$

Finding the normal reaction

M1

$$F = 0.4 \times 20 \times 9.8 \cos 30^\circ = 67.9 \text{ N}$$

Using $F = \mu R$

M1 A1

3

(iii) $T - 67.9 - 20 \times 9.8 \sin 30^\circ = 20 \times 3$

Four term equation of motion

Correct equation

M1 A1

$$T = 226 \text{ N}$$

Solving for T

Correct T

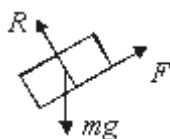
m1 A1

4

[11]

Q34.

(a)



Correct force diagram

B1

1

(b) $R = 7 \times 9.8 \cos 30^\circ = 59.4 \text{ N}$

Resolving perpendicular to plane

Correct force

M1/A1

2

(c) $F = 7 \times 9.8 \sin 30^\circ = 34.3 \text{ N}$

Resolving parallel to plane

Correct force

M1/A1

2

(d) $34.3 \leq \mu \times 59.4$

Using $F = \mu R$ or $F \leq \mu R$

M1

$$\mu \geq 0.577$$

Correct inequality

A1

2

- (e) Remains at rest, as mass appears in both F and R , so inequality unchanged.

Conclusion

B1

Reason

B1

2

[9]



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